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Chaos, Solitons and Fractals 17 (2003) 523–529

CHAOS
SOLITONS & FRACTALS

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An improved neural network based targeting method for chaotic dynamics

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Abstract

In this study, an improvement in the previously proposed Extended Control Regions (ECR) method [Physica D 150 (2001) 163] is presented, which gives better performance with respect to training time and average reaching time while maintaining the advantages of the previous method. In the improved approach, the system data is first pre-processed and subdivided into clusters, and then one artificial neural network is assigned to each such cluster; thereby the short-time-predictability property of chaotic systems is exploited more efficiently. The performances of different ECR versions have been compared on basis of simulation results obtained for several chaotic systems.

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1. Introduction

Since Ott et al. [1] have introduced a chaos control scheme (OGY method), the control of chaotic behaviour has gained much attention in many research areas. Then, several extensions and modifications of the OGY method have been proposed [2–9]. The most important advantage of the OGY method is the fact that it does not require any a priori knowledge about the system dynamics, whereas the major drawback of this method is the usually long reaching time until the system visits a close neighbourhood of the chosen target (either an unstable equilibrium point or an unstable periodic orbit of the system). Therefore, control of most chaotic systems via OGY method requires *targeting*, the task of steering the chaotic system from any initial point to a close neighbourhood of the target in the shortest possible time, by applying relatively small perturbations.

In literature, the first attempt to reduce the transient time was due to Shinbrot et al. [10–12]. This method has been extended to higher dimensions and refined via tree-type hierarchy in [13,14]. In [15], targeting is realised by creating periodic orbits without changing the topology of the system. Paskota et al. [16–19] have employed optimal control techniques for targeting purposes under some constraints. In [20], a mixed strategy is developed to make the controller more global. However, the methods proposed in [16–20] require the knowledge of the system dynamics, while the ones proposed in [21–23] do not require the knowledge about system behaviour. Iplikci and Denizhan [24,25] proposed the Extended Control Regions (ECR) method, where a single neural network is utilised for both targeting and local control tasks, and an improved version of the same method, respectively.

In this work, the ECR method has been further improved by clustering the activation regions into sub-regions and by using individual neural networks for each of these sub-regions. The last version of the ECR method gives better results compared to the previous versions with respect to training time and reaching time.

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2. The definition of control regions

The ECR method in all its versions maintains the attractive properties of the OGY method, i.e. no requirement of the a priori knowledge of the system dynamics and small control action via control parameter perturbation.

Since control of continuous-time chaotic systems with a periodic orbit as the target can be reduced to the following form, the ECR method will be presented for N -dimensional discrete-time chaotic systems with r control parameters and a target chosen as one of the unstable equilibrium points, only. Such a system can be represented by

$$\mathbf{z}_{n+1} = \mathbf{G}(\mathbf{z}_n, \mathbf{p}), \quad (1)$$

where $\mathbf{z}$ is the $N \times 1$ state vector, and $\mathbf{p}$ is the $r \times 1$ control parameter vector. Assume that this system has an equilibrium point at $\mathbf{z}^*$ described by

$$\mathbf{z}^* = \mathbf{G}(\mathbf{z}^*, \mathbf{p}_{\text{nom}}), \quad (2)$$

where $\mathbf{p}_{\text{nom}}$ is the $r \times 1$ vector of nominal parameter values. Each control variable (parameter) can be changed within its allowable range as follows:

$$\Pi = \{p^i : p_{\text{nom}}^i - \delta p_{\text{max}}^i < p^i < p_{\text{nom}}^i + \delta p_{\text{max}}^i, i = 1, \dots, r\}, \quad (3)$$

where δp_{max}^i is the maximum allowable parameter change for the i th parameter.

2.1. ECR-I

The main idea behind the ECR-I method is to identify different regions of the phase space, where the controller will be activated either for targeting or local (OGY) control purposes. These regions are referred to as the control regions ($\mathbf{S}_i, i = 0, 1, \dots, K$). Hence, the overall activation region of the controller is extended from $\mathbf{S}_0$ (as used in the OGY control) to the union of all $\mathbf{S}_i$'s ($i = 0, 1, \dots, K$). These control regions can be defined as follows:

Definition 1. If a system state $\mathbf{z}_n$ obeys (4) then $\mathbf{z}_n \in \mathbf{S}_0$.

$$\|\mathbf{z}_n - \mathbf{z}^*\| < \delta \quad \text{and} \quad \|\mathbf{G}(\mathbf{z}_n, \mathbf{p}) - \mathbf{z}^*\| < \delta \quad \text{with } \mathbf{p} \in \Pi. \quad (4)$$

Definition 2. $\mathbf{z}_n \in \mathbf{S}_1$ if it satisfies (5).

$$\mathbf{z}_n \notin \mathbf{S}_0 \quad \text{and} \quad \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n) \in \mathbf{S}_0 \quad \text{with } \mathbf{p}_n \in \Pi. \quad (5)$$

The definition can be generalised as follows: $\mathbf{z}_n \in \mathbf{S}_i$ if it satisfies (6).

$$\mathbf{z}_n \notin \mathbf{S}_0 \quad \text{and} \quad \mathbf{G}(\mathbf{G} \dots (\mathbf{G}(\mathbf{z}_n, \mathbf{p}_n) \mathbf{p}_{n+1}), \dots, \mathbf{p}_{n+i}) \in \mathbf{S}_0 \quad \text{with } \mathbf{p}_n, \dots, \mathbf{p}_{n+i} \in \Pi. \quad (6)$$

In other words, $\mathbf{z}_n \in \mathbf{S}_i$ if it is not in $\mathbf{S}_0$ but can be steered to $\mathbf{S}_0$ within i steps applying control parameters within the allowable range. If the dynamics of the system in $\mathbf{S}_i$'s is sufficiently modelled, the reaching time can be reduced at an increasing rate as K increases. However, any practical model of $\mathbf{S}_i$'s will usually fail to capture the fractal nature of the regions.

2.2. ECR-II

In order to avoid confusion, the control regions used in ECR-II will be referred to as $\mathbf{T}_i$'s ($i = 1, \dots, K$), which are defined in a slightly different manner than the $\mathbf{S}_i$ regions in ECR-I.

Definition 3. $\mathbf{z}_n \in \mathbf{T}_i$ if it satisfies (7).

$$\mathbf{z}_n \notin \mathbf{T}_{i-1} \quad \text{and} \quad \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n) \in \mathbf{T}_{i-1}, \quad \mathbf{p}_n \in \Pi, \quad i = 2, 3, \dots, K. \quad (7)$$

In other words, $\mathbf{z}_n \in \mathbf{T}_i$ if it is not in $\mathbf{T}_{i-1}$ but can be steered to $\mathbf{T}_{i-1}$ in one step applying control parameters within the allowable range. Obviously, $\mathbf{T}_i = \mathbf{S}_i$ for $i = 0, 1$.

2.3. ECR-III

Applications of the ECR approach to various chaotic systems have shown that control regions do not have to be connected or simply connected. Furthermore, because data corresponding to higher control regions tend to be scattered and finite amount of experimental data are used, sufficient information about the control regions may not be available.

The main idea in ECR-III is to try to extract more reliable models of system dynamics from the available data by considering data clusters separately instead of trying to interpolate between scattered region data.

Consequently, in ECR-III the control regions (T_i 's) as defined in ECR-II are clustered into smaller sub-regions (clusters) using a simple clustering algorithm, which will be explained in the following section. The system dynamics within each cluster of each control region is modelled separately. The loci of the clusters are represented in terms of the cluster means and the normalised cluster variances.

3. Application of ECR using neural networks

In all versions of the ECR method there is the need to model the relationship between a current state z_n , which is element of a control region and the corresponding parameter change p_n needed for targeting. In this study, this modelling is achieved via artificial neural networks, which are trained using system data. For that purpose RBF-based (Radial Basis Functions) neural networks have been preferred due to their better local approximation capability.

3.1. ECR-I

The training data gathered with random parameter variations within Π are classified into S_i regions according to Definitions 1 and 2, and then used to train a single neural network as described in Fig. 1.

Once the neural network is trained it can be used to obtain the necessary parameter change p_n corresponding to a current state z_n . An output of the neural network $p_n \notin \Pi$ is interpreted as an indicator of the fact that the current state is outside the control regions, hence the control parameters are left at their nominal values.

Since the targeting activity for all S_i regions is modelled by a single neural network, in ECR-I there is no need to know the actual region of the current state.

3.2. ECR-II

The approximation capacity of the single neural network employed in ECR-I does not allow too many control regions to be considered. Consequently, the most natural improvement of this method has been achieved by assigning a separate neural network to each control region T_i [25]. Such a division of tasks naturally results in a better modelling performance. In this case the training data belonging to any control region (T_i) have to be classified according to Definition 3 and used to train a neural network NN_i in a sequential manner ($i = 0, \dots, K$) (Figs. 2 and 3).

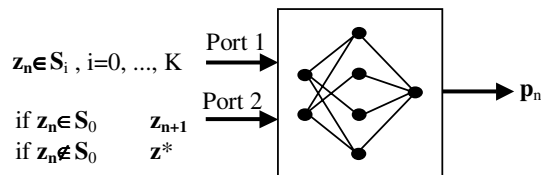


Fig. 1. Training phase of NN using ECR-I method.

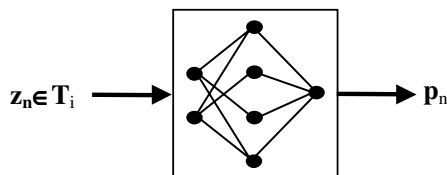


Fig. 2. Training phase of NN_i using ECR-II method.

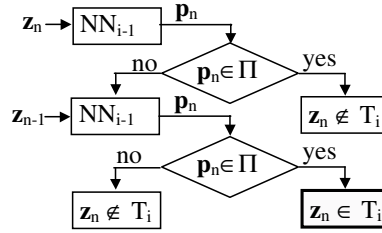


Fig. 3. Data extraction phase by Definition 3.

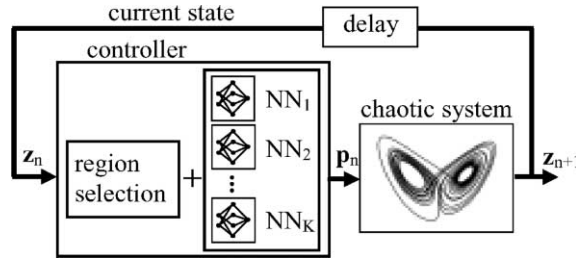


Fig. 4. The controller in the control loop in ECR-II.

Once all neural networks (NN_i , $i = 0, \dots, K$) have been trained, they can be used for targeting and local control as follows: The current state is fed sequentially to each neural network (NN_i , $i = 0, \dots, K$) starting from NN_0 , until one of them produces a $p_n \in \Pi$. The region with the smallest index, for which the neural network produces an acceptable parameter change, is regarded as the region of the current state. If none of the neural networks produces an acceptable parameter change, p_n is set to p_{nom} . The on-line control scheme is shown in Fig. 4.

3.3. ECR-III

In ECR-II it has been observed that the system data, in particular those belonging to higher control regions, can be quite scattered over the phase space. Consequently, the corresponding neural network tries to interpolate between these scattered data and eventually can provide wrong parameter changes. A remedy to this problem is proposed in ECR-III by identifying closer data groups by means of a simple clustering algorithm. The j th data cluster corresponding to the i th control region is denoted by C_{ij} and is assigned a neural network NN_{ij} . Each NN_{ij} is trained using the data corresponding to C_{ij} to provide the necessary control parameter values that will take the system to $C_{(i-1)k}$, a cluster of the $(i-1)$ st control region at the next step. The simple clustering algorithm used for determining any C_{ij} can be explained as follows:

An arbitrary data point in a control region T_i is taken as a centre of a hyper-sphere with a radius r , which is chosen as the first minimum of the histogram of the inter-data distances by visual analysis.

First a radius r is chosen as the first minimum of the histogram of the inter-data distances by visual analysis. Data points, which lie within r -neighbourhood of each other, are assigned to the same cluster.

After the clusters, C_{ij} are obtained, the *Normalized Covariance Matrix* of the cluster C_{ij} denoted by $\underline{N}_{ij}$ and the mean of cluster C_{ij} denoted by μ_{ij} are calculated. This modification introduced in ECR-III allows the determination of the current state location in an analytical manner as follows:

For the current state z_n , the *Normalized Mahalanobis Distance* (NMD) from all C_{ij} 's are calculated using Eq. (8).

$$(\text{NMD})_{ij}^2(z_n) = (z_n - \mu_{ij})^T \underline{N}_{ij} (z_n - \mu_{ij}), \quad (8)$$

where $\underline{N}_{ij}$ is the Normalized Covariance Matrix and μ_{ij} is the mean of C_{ij} .

The current state is assumed to be in the cluster C_{ij} , to which it has the minimum Normalised Mahalanobis Distance NMD_{ij} . In this case the corresponding neural network NN_{ij} is activated to obtain the necessary control parameter(s) for targeting. If NN_{ij} produces a $p_n \notin \Pi$, no targeting is applied.

Table 1

Comparison of the methods with respect to training time and average reaching time

	OGY	ECR-I	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR- III	ECR-II	ECR-III	ECR-II	ECR-III
<i>Logistic map</i>														
# of activation regions	0	2	1		2		3		4		5		6	
Training time (flops)	0.16	35	2.89	3.56	4.05	3.73	5.33	3.96	6.72	4.07	6.76	4.18	8.06	4.21
	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$
Av. reach. t. (noiseless)	143.7	50.87	51.31	58.571	46.37	53.027	13.75	8.575	9.32	5.753	9.23	5.316	9.14	5.523
Av. reach. t. (0.30% noise)	156.8	52.64	52.96	58.153	46.99	52.797	14.16	8.694	10.96	5.894	11.26	5.418	11.25	5.747
<i>Henon map</i>														
# of activation regions	0	3	3		4		5		6		7		8	
Training time (flops)	0.92	601	33	28.4	35	29.2	42	29.8	48	30.5	56	31.0	57	31.6
	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$
Av. reach. t. (noiseless)	767.9	108.73	107.18	121.76	101.72	89.56	59.89	57.95	30.06	27.89	21.94	17.17	22.50	15.39
Av. reach. t. (0.22% noise)	837.8	134.9	126.00	108.58	104.82	82.56	60.25	57.59	39.29	26.52	31.00	15.51	30.65	13.64
<i>Lorenz system (real coord.)</i>														
# of activation regions	0	1	1		2		3		4		5		6	
Training time (flops)	0.21	96	443	49.9	454	50.6	455	51.1	455	51.4	456	51.5	456	51.5
	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$	$\times 10^6$
Av. reach. t. (noiseless)	15.09	6.64	6.14	6.80	6.43	6.44	6.45	6.43	6.57	6.39	6.45	6.42	6.66	6.42
Av. reach. t. (0.08% noise)	18.91	6.81	6.43	7.93	6.02	7.79	6.34	7.49	5.93	6.82	5.92	6.65	6.86	6.59
<i>Lorenz system (delay coord.)</i>														
# of activation regions	0	2	1		2		3		4		5		6	
Training time (flops)	0.24	5.91	4.81	0.84	4.84	0.87	4.86	0.88	4.87	0.89	4.88	0.89	4.89	0.89
	$\times 10^6$	$\times 10^{10}$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$
Av. reach. t. (noiseless)	35.32	18.31	17.92	16.21	13.62	14.90	12.83	14.02	12.76	12.23	12.73	12.61	12.73	12.47
Av. reach. t. (0.36% noise)	34.20	20.08	18.93	15.37	15.31	15.48	14.21	12.94	15.24	11.67	15.26	11.70	15.22	11.25
<i>Duffing oscillator (delay coord.)</i>														
# of activation regions	0	2	1		2		3		4		5		6	
Training time (flops)	0.48	5.03	1.08	0.25	1.12	0.26	1.14	0.26	1.15	0.26	1.15	0.26	1.16	0.27
	$\times 10^6$	$\times 10^{10}$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$	$\times 10^9$
Av. reach. t. (noiseless)	107.55	42.94	39.81	42.86	16.40	21.64	13.20	11.23	13.98	11.56	14.03	12.91	13.68	15.47
Av. reach. t. (0.27% noise)	116.70	48.40	51.40	60.67	27.32	36.10	18.65	17.64	18.97	17.40	18.47	21.61	17.83	23.40

4. Simulation results obtained for various chaotic systems

Simulations have been performed on various chaotic systems for both under noiseless and noisy conditions, where some random noise proportional to the root mean square value of the measured data is added. The average reaching time is calculated as the average of the time it takes the system to enter the OGY region starting from many different initial conditions. Table 1 summarises the performances of OGY, ECR-I, ECR-II and ECR-III methods for different chaotic systems given in Table 2.

5. Conclusions

In this work, the three different versions of the neural network based ECR method have been presented. These versions differ from each other in the way extended control regions are defined and modelled, and the way the region of a current system state is determined. The simulation results show that ECR-III gives the best performance both in terms of average reaching time and off-line training time of the neural networks as a result of splitting the region data into sub-clusters. In ECR-III a very simple clustering algorithm has been preferred although more sophisticated clustering algorithms may improve the average reaching time even further on cost of increasing the training time.

Acknowledgements

This research has been completed at the Mechatronics Research and Application Centre of Bogazici University and supported by the Research Fund of Bogazici University, Project no. 00A204.

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